

Yanjin Li

Self-taught AI Engineer & Researcher — research · systems · applied ML · product

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Self-taught AI builder — learning from papers and courses since early 2026, I designed, built and evaluated **11 projects in three months**, spanning training-method research, multi-device AI systems, applied decision systems and shipped products.

SELECTED PROJECTS

Each title links to its full technical write-up at yanjin-ai.github.io/portfolio.

Zero Gradient ↗ — a 4.16B-parameter content-routed MoE language model trained with backpropagation globally disabled, on a single 16GB T4 RESEARCH

- Replaced end-to-end gradients with layer-local losses + depth-supervised readout heads, EMA-prototype content routing and frozen attention; 4 layers × 950 experts, top-2 routing, only **0.42% of experts updated per step** — fully deterministic and reproducible.
- Ran a 5-phase study; reached **test ppl 1355** vs 159 for a same-budget backprop control (the measured 3–9× cost). Sentiment transfers at 79%, but relational (NLI) and arithmetic stay at chance.
- Falsified my own 3 mechanistic hypotheses and corrected 2 false positives; a controlled trainable-attention transformer solved the same tasks at 100% (69.97% on real SNLI), **proving the ceiling is architectural, not the absence of gradients**. Paper draft + 8-page workshop paper + arXiv package.

[PyTorch \(autograd off\)](#) · [MoE 950 experts/layer](#) · [layer-local learning](#) · [EMA routing](#) · [WikiText-103](#) · [Kaggle T4](#)

Gemma4all ↗ — a cross-device AI task system: a phone task is routed to phone / desktop / cloud by privacy & capability, hands off across devices, and every state change is an immutable event SYSTEMS

- Four principles — task-first, schema-first, control-plane-first, local-first; **42 JSON Schemas generate all types** (zero hand-written DTOs across four languages), with 7 protocol docs + ADRs.
- Built a TypeScript/Node control plane (event-sourced, SQLite, AJV validation, 7 API groups), a Python asyncio desktop runtime with 6 real tools + 16 tests, and native iOS/Android/macOS hosts discovering over Bonjour; on-device inference via **LiteRT-LM on Metal GPU** (4–6s/task).
- A privacy-based routing engine + idempotent cross-device handoff; a dedicated security phase added a central policy, catastrophic-pattern screening, a **fail-closed human-in-the-loop approval gate**, and CI.

[TypeScript](#) · [Python](#) · [Swift](#) · [Kotlin](#) · [JSON Schema 2020-12](#) · [LiteRT-LM / Metal](#) · [SQLite](#) · [event sourcing](#) · [Bonjour](#)

TriageGeist ↗ — an emergency-department triage decision stack that chose data honesty over leaderboard rank APPLIED ML

- Discovered **99.7% of chief-complaint phrases mapped monotonically to one triage label** (a fusion model hit QWK≈1.0) and refused to exploit it — documented it as a data-quality finding.
- Built a text-blind multi-task GBDT instead (acuity + admission + length-of-stay, **QWK 0.9287**) on physiology features via one `transform()` shared by train and serve (no skew).
- Wrapped it in governance: isotonic calibration (**ECE –46%**), conformal prediction (defers ~89% of cases to human review), bootstrap fairness CIs (found a 3× undertriage gap for Estonian speakers), an error taxonomy and a 4-view ops console. **12/12 tests**.

[LightGBM \(multi-task\)](#) · [isotonic calibration](#) · [conformal prediction \(APS\)](#) · [bootstrap fairness](#) · [shared transform\(\)](#)

NeuroGolf (ARC-AGI) ↗ — compiling each of 400 ARC-AGI tasks into a minimal static ONNX network (scored $\max(1, 25 - \ln(\text{params} + \text{memory}))$) RESEARCH

- Reframed a "neural golf" contest from a learning problem into **program synthesis + circuit golfing**, realizing the official per-task generator is the ground-truth rule and same-distribution sampler.
- Reverse-engineered the scorer into a cost model (input/output & node attributes free; single-node circuits ≈ 0 memory; log-scale ⇒ redesign » tuning); built an oracle, a graph library, and a 3-level validator (official + **3000+ generated samples** + constraint audit).
- Key lesson: an LP-fit conv passed 3000 local samples but **scored 0 on Kaggle** — distinguishing "fitting a function that passes the samples" from "implementing the true rule"; only integer-weight circuits (margin ≥1) are private-set-robust. Wrote a transferable methodology note.

[ONNX / onnxruntime](#) · [program synthesis](#) · [scorer reverse-engineering](#) · [integer-weight circuits](#) · [LP / torch fitting](#)

AI Showrunner — one sentence → a subtitled 9:16 short film; an end-to-end AI virtual-production pipeline PRODUCT

- Pipeline: script → storyboard → per-shot video generation → a vision-model critic → multi-language subtitles, all as **replayable Pydantic JSON artifacts** with human-in-the-loop gates and per-run cost ceilings.
- A **closed-loop critic** (not one-shot): each clip is sampled at 3 frames and scored by a VL model on narrative alignment / character consistency / technical quality; failures produce targeted fix-prompts within a retry budget.
- Cross-shot character consistency via canonical reference portraits injected as image-to-video first frames (verified 9–10); **six complete end-to-end films** produced.

FastAPI · asyncio · Pydantic · Qwen-Max planner · Qwen-VL critic · Wan I2V/T2V · ffmpeg

Nemotron Post-training — a reproducible post-training pipeline for NVIDIA Nemotron-3-Nano-30B (MoE + Mamba-Transformer hybrid); a fully root-caused failure RESEARCH

- Built a rule-teacher → CoT-SFT pipeline: a deterministic rule solver as teacher (**EM 0.75**, 100% format) for clean distillation labels; loaded the 30B MoE in **4-bit on one GPU**; found and patched a `cache_position` bug; masked SFT collator; multi-dimensional reward shaping for GRPO/DPO.
- SFT loss fell to 0.33 but generation collapsed (EM 0%); **root-caused the teacher-forcing / generation mismatch**, wrote six postmortems, and designed the native-PEFT recovery route.
- Distilled a permanent process rule: sanity-check the full train→generate loop on 5 samples before any long run.

Transformers · PEFT/LoRA · Unsloth · TRL · 4-bit quantization · GRPO reward shaping

Also built: Offline French Companion (in-browser LLM tutor — WebLLM + Whisper + OCR, fully offline) · Orbit Wars (multi-agent evaluation methodology) · Maze Crawler (strategy agent with CI) · VoiceInput (pure-Swift macOS speech input) · Atelier Post (shipped art-to-Instagram tool, web/PWA/iOS — live).

PROFESSIONAL EXPERIENCE

BNP Paribas — Global Corporate Banking Intern Paris · Dec 2025 — Present

- Structured tailored cross-border transactional and financing solutions for multinational clients; built and presented pitchbooks combining market analysis with pricing and cross-border liquidity.
- Helped land internal AI tooling — built RAG knowledge bases on Open WebUI and Dify to improve client-response efficiency (book incl. CATL, BYD, Alipay, Midea, TCL, JD.com).

HSBC — Global Markets, Liquidity & Investment Intern Paris · Jun — Nov 2024

- Monitored Fed/ECB/BoE rates and money markets to optimize short-term EMEA portfolios under LCR/NSFR; built automated VBA/Excel tools tracking rate moves and pricing gaps in real time.
- Developed dynamic pricing models for wholesale deposits; contributed to the ALCM committee (gap-risk analysis, funding-cost optimization).

TikTok — Growth Strategy Intern Beijing · Feb — May 2023

- Contributed to 10+ strategic projects, growing SaaS/AI market share by 23%; ran SQL/Excel analytics for a 20+ member team and lifted user conversion by 13% in one month.

Uber China · DiDi — Product Management & User Operations Intern Beijing · Aug 2022 — Jan 2023

- Managed 3 business-operations projects; analyzed large datasets in Python, SQL and VBA to build predictive models and data-driven user-interaction strategies.

EDUCATION

ESCP Business School (Paris & London) — Master in Management, Finance 2023 — 2026

Beijing International Studies University — BA Language & Culture (French, Japanese); minor New Media (CUC); self-taught Psychology (Yale/Coursera)

2019 — 2023

SKILLS

Programming: Python · TypeScript/Node · Swift · Kotlin · JavaScript · SQL · VBA

AI / ML: PyTorch · LoRA/PEFT · MoE · calibration · conformal prediction · LiteRT-LM · ONNX · WebLLM · Whisper · LLM/agent orchestration · schema-first & event-sourced systems

Spoken languages: Chinese (native) · English (fluent) · French (fluent) · Japanese (working)

Strengths: High agency and a self-taught, zero-to-one builder — curious, reflective and resilient, with strong insight and a painter's eye for design, at home across cultures and languages.